Q. Convert the Grammar into GNF (Greibach Normal Form)
(7 Marks)

$S \to ABb \mid a$

$A \to aaA \mid B$

$B \to bAb$

G/CFG $\to$ GNF

⇒ Step 1: Eliminate Null Productions

→ in this example, there are no null productions.

Step 2: Elimination of Unit Productions (i.e. single capital letter)

→ in this example, $A \to B$ is a unit production.

∴ Replace A with production of B

Since $B \to bAb$, substitute this into A

∴ $A \to aaA \mid bAb$

Now, remove $A \to B$

Grammar after eliminating unit productions:

$S \to ABb \mid a$

$A \to aaA \mid bAb$

$B \to bAb$

Step 3: obtain GNF

→ identify GNF productions

- write GNF productions as it is which are already in GNF.

Condition for GNF: Every production must begin with a terminal: $A \to a\alpha$

where: a is a terminal

α is a sequence of non terminals

→ in this example, only one production is in GNF form.

i.e. $S \to a$

$S \rightarrow ABb \mid a$
$A \rightarrow aaA \mid bAb$
$B \rightarrow bAb$

→ $S \rightarrow aaABb \mid bAbBb \mid a$
$A \rightarrow aaA \mid bAb$
$B \rightarrow bAb$

replaced productions of A in $S \rightarrow ABb$

- Since productio $S \rightarrow ABb$ starts with non-terminal A, substitute productions of A, so that S begins with terminals.

→ $S \rightarrow aCABD \mid bADBD \mid a$
$C \rightarrow a$
$D \rightarrow b$
$A \rightarrow aCA \mid bAD$
$B \rightarrow bAD$

converted terminal symbols into non-terminals in production S. ⇒ $C \rightarrow a$, $D \rightarrow b$
(because every production should starts with a terminal symbol followed by non-terminals only)

- In this, we introduced new variables such as C & D for terminals appearing after the first position because in GNF, only first symbol can be a terminal

* Now every production begins with a terminal followed by only non-terminals. Hence below given grammar is in GNF.

→ $S \rightarrow aCABD \mid bADBD \mid a$
$A \rightarrow aCA \mid bAD$
$B \rightarrow bAD$
$C \rightarrow a$
$D \rightarrow b$

GNF form (Answer)

Q. Apply the rules and show step by step conversion of the following grammar to CNF. (7 Marks)

$S \rightarrow ASA \mid aB$

$A \rightarrow B \mid S$

$B \rightarrow b \mid \epsilon$

G/CFG → CNF

⟹

In CNF, on the right side of CFG, only two types of length is allowed.

(i) length 1 (no of symbol is 1) : terminal symbol

(ii) length 2 (no of symbol is 2) : both should be non terminal

i.e. A production of form :

$A \rightarrow BC$ or $A \rightarrow a$ } → Condition for CNF

where $A, B, C \in V$ and $a \in T$, is called as CNF.

Here, V = non terminal symbols (capital letters)

T = terminal symbol (small letters, special symbols)

Step 1 : Eliminate Null Productions

put ε in place of B in every production.

$S \rightarrow ASA \mid aB \mid a$

$A \rightarrow B \mid S \mid \epsilon$

$B \rightarrow b$

⟹ Now put $A = \epsilon$ in the productions containing A

$S \rightarrow ASA \mid SA \mid AS \mid S \mid aB \mid a$

$A \rightarrow B \mid S$

$B \rightarrow b$

Step 2 : Elimination of Unit Productions (i.e. single capital letter)

$A \rightarrow B$

$A \rightarrow S$

$S \rightarrow S$

→ Remove $S \rightarrow S$

→ Replace $A \rightarrow B$

since $B \rightarrow b$,

add $A \rightarrow b$ ✓

→ Replace $A \rightarrow S$

Copy productions of S : ∴ $A \rightarrow ASA \mid SA \mid AS \mid aB \mid a$ ✓

→ Grammar becomes:

$S \to ASA \mid SA \mid AS \mid aB \mid a$
$A \to ASA \mid SA \mid AS \mid aB \mid a \mid b$
$B \to b$

Step 3: Replace terminals in mixed productions

mixed productions means terminal + non-terminal

→ in this example, mixed production is aB

∴ Introduce new Non terminal for the terminal 'a'

$C \to a$

Replace: $aB \to CB$

∴ Grammar becomes:

$S \to ASA \mid SA \mid AS \mid CB \mid a$
$A \to ASA \mid SA \mid AS \mid CB \mid a \mid b$
$B \to b$
$C \to a$

Step 4: Convert long productions into 2 variables/non-terminals

→ in this example, long production is $S \to ASA$

we need to convert it into 2 non-terminals.

- for this, keep first non-terminal as it is, and for other two non-terminals, introduce new non-terminal.

$S \to AD$

here, $D \to SA$ [D is the new non-terminal]

$$\left.\begin{aligned} S &\to AD \mid SA \mid AS \mid CB \mid a \\ A &\to AD \mid SA \mid AS \mid CB \mid a \mid b \\ D &\to SA \\ B &\to b \\ C &\to a \end{aligned}\right\} \to$$

as every production is of the form: $A \to BC$ or $A \to a$, this grammar is in CNF

(Answer)

Grammar | CFG → CNF PYQs

Q. Convert the following grammar into CNF (7 Marks)

$S \to cBA$
$S \to A$
$A \to cB$
$A \to AbbS$
$B \to aaaa$

Q. Convert the CFG, $G(\{S, A, B\}, \{a, b\}, P, S)$ to CNF, where P is as follows : (7 Marks)

$S \to aAbB$
$A \to Ab \mid b$
$B \to Ba \mid a$

Q. Given the context-free grammar G, find a CFG G' in Chomsky Normal Form. (7 Marks)

$S \to AaA \mid CA \mid BaB$
$A \to aaBa \mid DC$
$B \to bb \mid aS$
$C \to Ca \mid bC \mid D$
$D \to bD \mid \Lambda$

Q. Convert the following grammar to CNF

$S \to ABA$ (7 Marks)
$A \to aA \mid \epsilon$
$B \to bB \mid \epsilon$

Q. Apply the rules and show step by step conversion of the following grammar to CNF

$S \to ABCBCDA$ (7 Marks)
$A \to CD$
$B \to Cb$
$C \to a \mid \lambda$
$D \to bD \mid \lambda$

Q. Convert following CFG to CNF (7 Marks)

$S \rightarrow S(S) \mid \wedge$

Q. Convert following CFG to CNF (7 Marks)

$S \rightarrow aX \mid Yb$

$X \rightarrow S \mid \wedge$

$Y \rightarrow bY \mid b$

Q. Define Chomsky Normal Form (CNF). Convert the following CFG into its equivalent CNF.

$S \rightarrow aY \mid Ybb \mid Y$ (7 Marks)

$X \rightarrow \wedge \mid a$

$Y \rightarrow aXY \mid bb \mid XXa$

Q. Define: CNF. Show the steps to convert CFG into CNF. Convert the following CFG into equivalent CNF.

$S \rightarrow TU$ (7 Marks)

$T \rightarrow 0T1 \mid \epsilon$

$U \rightarrow 1U0 \mid \epsilon$

Q. Convert the following grammar into CNF :

$S \rightarrow cBA$ (7 Marks)

$S \rightarrow A$

$A \rightarrow cB$

$A \rightarrow AbbS$

$B \rightarrow aaaa$